



Physiologically based pharmacokinetic (PBPK) modeling of perfluorohexane sulfonate (PFHxS) in humans

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ABSTRACT

Per- and polyfluoroalkyl substances (PFAS) are persistent, man-made compounds prevalent in the environment and consistently identified in human biomonitoring samples. In particular, perfluorooctane sulfonic acid (PFOS), perfluorooctanoic acid (PFOA), and perfluorohexane sulfonic acid (PFHxS) have been identified at U.S. Air Force installations. The study of human toxicokinetics and physiologically based pharmacokinetic (PBPK) modeling of PFHxS has been less robust and has been limited in scope and application as compared to PFOS and PFOA. The primary goal of the current effort was to develop a PBPK model describing PFHxS disposition in humans that can be applied to retrospective, current, and future human health risk assessment of PFHxS. An existing model developed for PFOS and PFOA was modified and key parameter values for exposure and toxicokinetics were calibrated for PFHxS prediction based on human biomonitoring data, particularly general population serum levels from the U.S. Centers for Disease Prevention and Control (CDC) National Health and Nutrition Examination Survey (NHANES). Agreement between the model and the calibration and evaluation data was excellent and recapitulated observed trends across sex, age, and calendar years. Confidence in the model is greatest for application to adults in the 2000–2018 time frame and for shorter-term future projections.

1. Introduction

Per- and polyfluoroalkyl substances (PFAS) are persistent, man-made compounds prevalent in the environment and consistently identified in human biomonitoring samples (CDC, 2021; East et al., 2021). In particular, perfluorooctane sulfonic acid (PFOS), perfluorooctanoic acid (PFOA), and perfluorohexane sulfonic acid (PFHxS) have been identified at U.S. Air Force installations. As reported by East et al. (2021), “We applied a data science approach to characterize and prioritize PFAS and PFAS mixtures from a large dataset of PFAS measurements in surface waters associated with US Air Force Installations with a history of the use of aqueous film-forming foams (AFFFs). Several iterations of stakeholder feedback culminated in a few main points [...] First, perfluorooctane sulfonate (PFOS) was often a dominant PFAS in a given surface water sample, frequently followed by perfluorohexane sulfonate (PFHxS). Second, a 4-chemical mixture generally accounted for >80% of the sum of all routinely reported PFAS in a sample, and the most representative 4-chemical mixture was composed of PFOS, PFHxS, perfluorohexanoic acid (PFHxA), and perfluorooctanoic acid (PFOA).”

Associations between PFHxS plasma/serum levels and health effect biomarkers have been noted in cross-sectional studies of the general

population. Endpoints identified in these studies as associated with serum PFHxS level included increased androgens (a risk factor for cardiometabolic diseases) in overweight/obese postmenopausal women (Wang et al., 2021), altered thyroid function (free T4 levels) in non-smokers (van Gerwen et al., 2020), and increased inflammatory and oxidative stress markers (increased lymphocyte counts and increased serum iron, albumin, and total bilirubin) (Omoike et al., 2020). However, associations may be manifestations of reverse causality (Andersen et al., 2021). For example, changes in renal function (Jain and Ducatman, 2019) may produce changes in plasma PFAS concentration, rather than the other way around. Furthermore, in these cross-sectional studies, attribution of causality to PFHxS may be spurious due to potential correlation with exposure to other chemicals (especially other PFAS) or potency overestimated due to co-exposure to other chemical classes with potential to contribute to the same observed effect.

The toxicokinetics of PFOS and PFOA have been extensively studied and modeled in support of human health risk assessment (e.g., Loccisano et al., 2011; Worley et al., 2017; Chou and Lin 2019, 2020; Ruark et al., 2017; Deepika et al., 2021). The model structures are very similar, consisting of a plasma compartment and distribution of unbound compound via plasma flow into several tissue compartments. The chemical

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was assumed to be cleared by way of renal filtration of unbound compound, with potential reuptake into the kidney mediated via organic anion transporting peptides (OATPs) (Han et al., 2012). The earlier model of Andersen et al. (2021) also incorporated the element of saturable reuptake, but was a simpler model with fewer compartments.

In contrast, study of human toxicokinetics and physiologically based pharmacokinetic (PBPK) modeling of PFHxS has been less robust and limited in scope and application, and has relied on models with structures similar to those employed for PFOS and PFOA (Fabrega et al., 2015; Kim et al., 2018). With respect to general kinetics of PFHxS in rats, shorter plasma half lives were observed for females as compared to males in at least three separate studies (Sundstrom et al., 2012; Huang et al., 2019; Kim et al., 2016). That finding is consistent with other PFAS (PFOA, in particular) and the difference has often been attributed to differences in abundance of transporters responsible for renal resorption of PFAS by OATPs (Kudo et al., 2002). PFOS and PFHxS demonstrated similar KM values for human Na⁺/taurocholate cotransporting peptide (a hepatocyte bile acid transporter) when expressed in CHO cells (Zhao et al., 2015). Three human OATPs and 3 rat OATPs also displayed uptake capabilities for PFBS, PFHxS, and PFOS (Zhao et al., 2017). No studies specifically addressing kidney transporter uptake of PFHxS were identified.

A more rigorously evaluated PBPK model for human PFHxS disposition that incorporates historical trends in PFHxS exposure will support human health risk assessment where potential effects of PFHxS are of concern, whether alone or more likely as part of a mixture (East et al., 2021). The primary goal of the current effort was to develop a human PBPK model for PFHxS disposition in adults that can be applied to retrospective, current, and future human health risk assessment of PFHxS in U.S. adults. In addition, the strategies used to develop and evaluate this model have the potential to serve as a template for other human PFAS PBPK models with similar data availability.

2. Methods

2.1. Model development strategy

2.1.1. Overview

The general workflow toward using PBPK models for human health risk assessment involving PFHxS exposure involved three phases: scoping, model refinement and evaluation, and application. These phases are described in additional detail in the following paragraphs and the Supplementary Material.

2.1.2. Scoping phase

The scoping phase included review of existing human PBPK models for PFHxS and other PFAS, review and analysis of human PFHxS biomonitoring data sets; identification of needed/desirable features for the PBPK model and key data sets; and selection of base model(s) for evaluation and refinement. These scoping activities are described in greater detail in the Supplementary Material. Briefly, PFHxS-specific models (Fabrega et al., 2015; Kim et al., 2018) were found to be similar in structure to each other and other PFAS models while differing in observed plasma clearance behavior. A PFOS and PFOA lifestage PBPK model incorporating age and calendar year dependent exposure (Ruark et al., 2017) was selected as the base model for subsequent model refinement, evaluation, and application.

2.1.3. Model refinement and evaluation phase

Model refinement and evaluation started with efforts to reproduce previous model simulations (Fabrega et al., 2015; Kim et al., 2018). Next, preliminary assignment of data sets for PFHxS elimination calibration was made. A single sensitive toxicokinetic parameter value (transporter-mediated reuptake of PFHxS into the kidney) was adjusted to fit the half-life data of Li et al. (2018). Calibration of the body weight normalized maximal reabsorption rate (TMC) was chosen as the

approach to matching experimentally observed half-lives rather than adjustment of affinity (KT) based on an assumption of a sex-specific difference in transporter abundance (relevant to TMC), not affinity (relevant to KT) (Kumar et al., 2021). These steps in the model refinement are described in greater detail in the Supplementary Material.

With the toxicokinetic parameter values set, an initial calibration of the baseline U.S. intake rate and calendar year specific exposure pattern was conducted using gender- and age-group specific median plasma PFHxS data for 1999–2018 from the National Health and Nutrition Examination Survey (NHANES) (CDC, 2019/2020) and 1974 and 1989 biomonitoring data from Olsen et al. (2005). A preliminary set of sensitivity analyses were conducted next, as described below. Upon review of these initial sensitivity analyses, an adjustment factor for childhood intake for ages 0–10 years old (Noorlander et al., 2011; Ruark et al., 2017; Deepika et al., 2021) was calibrated. Body weight-normalized daily intake of PFAS may be elevated in children due to greater ingestion of house dust and greater dermal contact with PFAS-containing consumer products (Winkens et al., 2017). Calendar year trends for Europe were also evaluated and reported in the Supplementary Material.

After finalizing the baseline model parameterization for U.S. populations, revised sensitivity analyses were conducted using the updated parameterization. Additional representative U.S. serum or plasma data sets for general environmental exposure were simulated, with or without calibration of subpopulation-specific intake rates, but no further adjustment of toxicokinetic parameters or calendar year-specific U.S. trends was made. Interindividual variability identifiable from retired U.S. fluorochemical worker plasma data was characterized (Olsen et al., 2007) by adjusting additional exposure above background during their employment period and the rate of transporter-mediated reuptake of PFHxS. The relationship between plasma or serum PFHxS level and predicted urinary PFHxS concentration was evaluated primarily based on the occupational exposure data of Fu et al. (2016).

2.1.4. Model application phase

The initial demonstration of the PBPK model utility in relation to risk assessment consisted of two activities. First, two instances from the literature in which an external (in vivo) exposure of PFHxS was related to an internal dose associated with a health effect were modeled (Ali et al., 2019; EFSA, 2020). External doses that these authors determined to be equivalent to their health-based internal (plasma) levels of concern were used as the adult daily exposure rate in the model reported here. Predictions from the newly developed model were compared to the internal point of departure for these two previous health based guidance value efforts. Second, derivations of time-to-target internal dose were made for adult blood plasma PFHxS concentrations of 5 ng/ml, after a step increase in PFHxS intake. The use of this value as a target is not intended to imply that this value is an appropriate health-based guideline for PFHxS. However, some authorities have established health-based PFAS guidelines based on serum PFOS, PFOA, or the sum of specific PFAS components at levels in this range (Mikkonen et al., 2021). The target of 5 ng/ml was selected based on analogy to German PFOA and PFOS biomonitoring assessments. Schumann et al. (2021) derived human biomonitoring (HBM)-I values (no action needed if the value is not exceeded) of 2 ng/ml for PFOA and 5 ng/ml for PFOS. The lowest derived HBM-II value (a concentration which, “when exceeded, may lead to health impairment”) was 5 ng/ml PFOA in women of child-bearing age, with higher HBM-II values for other subpopulations and for PFOS. Thus a 5 ng/ml target for PFHxS would generally be expected to be health protective but with an unknown margin of safety. The time-to-target derivations for PFHxS assumed only background PFHxS exposure up to January 1, 2022. After that date, background intake was assumed to continue, but additional PFHxS, at a constant mg/kg body weight/day intake rate was assumed. Both males and females were simulated. Starting ages (as of January 1, 2022) of 18, 25, 35, 45, 55, and 65 years were considered and predictions were not extended beyond

age 80 years for greatest confidence in estimation of the physiological (Ruark et al., 2017) and chemical-specific pharmacokinetic parameters. Two intake levels for lifetime background exposure were simulated: a central tendency rate and a high rate. The central tendency rate for the time-to-target simulations was the same rate derived for predicting the NHANES data; the high rate was the central tendency rate multiplied by a factor of 5, to reflect (slightly exceed) the approximate ratio of the 95th percentile NHANES values to the medians (CDC 2019, 2020).

2.2. Data analysis

2.2.1. Data extraction

Tabular data and values mentioned in the text were used when available. When information was available only via published figures, estimates were made by digitizing data (symbols or portions of lines depicting simulations) figures by using WebPlotDigitizer v. 4.4 (November 2020) or earlier versions (<https://automeris.io/WebPlotDigitizer>, by Ankit Rohatgi, Pacifica, California).

2.2.2. Treatment of National Health and Nutrition Examination Survey (NHANES) data

Data sets of PFAS measurements and subject demographic information were downloaded from the CDC website (CDC, 2019, 2020). For modeling analysis, the data from each wave (two-year period) was grouped by gender and by age (12–15, 16–30, 31–45, 46–60, and 61 years and up). For each wave, gender, and age group, the following statistics were derived: the number of plasma samples, percentage of samples with PFHxS above the limit of detection, and the following features of the serum PFHxS concentration distribution: the weighted average; 25th, 50th (median), 75th and 95th percentiles; and the maximum. NHANES sampling is not random, but oversamples certain groups (Chen et al., 2020). The weighted average and percentile values help adjust for the nonrandom design. The age groups were selected to be consistent with the analysis by Toms et al. (2019) and because these divisions were generally consistent with age-related trends observed in the data (see Supplementary Materials, Figure S-3). PBPK model predictions for the approximate age midpoints (ages 14, 23, 38, 53, and 70 years old) were compared to the medians of the grouped NHANES data. Data for each two-year wave were assumed to have been collected at the midpoint of the period (e.g., 1999–2000 data were assumed to have been collected January 1, 2000). For the three most recent NHANES waves, data for pregnant women were identified and then compared to data for all women from the same age group and wave, due to concerns about early-life PFAS exposures EFSA (2020).

2.2.3. Treatment of other U.S. data used for model calibration and evaluation

Descriptions of data for Olsen et al. (2005, 2007), Wang et al. (2011), and Fu et al. (2016) are provided in detail in the Supplementary Materials, section S.1.2.5.

2.3. Simulation, model optimization, and evaluation of agreement between models and predictions

All simulations were conducted using acslX (AEGIS Technologies, version 3.1.5.1) using Gear's algorithm. The model csl code (found in the Supplementary Material) was modified from what was provided in the supplementary material of Ruark et al. (2017). The modifications were limited to adding PFHxS-specific values for exposure and toxicokinetic parameters, adding flexibility and complexity to calendar-year dependent exposure; and adding an exposure pathway distinct from background. This additional pathway was used to describe occupational exposure (Olsen et al., 2007) and for model application to "time to target" internal dose predictions for a hypothetical group of adults with new additional PFHxS exposure. Scripts for making scenario-specific predictions described in this work and for parameter

estimation were also provided in the Supplementary Material.

Tests for statistically significant improvements in fit for a two-parameter optimization (2 degrees of freedom [df]) as compared to optimization of one parameter (1 df) were applied to the model evaluation of the Olsen et al. (2007) data using the method of Collins et al. (1999). Optimizations were conducted using the built-in parameter estimation routines in acslX, applying the Nelder-Mead method with heteroscedasticity fixed at a value of two (relative error method), with the Log Likelihood Function as the metric to be optimized.

Quality of model fit (agreement between predictions and measured values) was assessed as average fold error (AFE) and average absolute fold error (AAFE) of PFHxS concentration. AFE was calculated as geometric mean of the ratio of the predicted/measured value (measured values were typically reported as a median or geometric mean) for each point in the time series and can hypothetically range from zero to infinity. An AFE of 1 indicates that model predictions are balanced with respect to over-vs. underprediction of the data. AAFE was calculated for the ratios of absolute error, which is the greater of predicted/measure value or measured/predicted value, and thus must be greater than one. Perfect agreement between the measured and predicted values would be indicated by $AAFE = 1$. To be of sufficient quality for use in risk assessment, the International Programme on Chemical Safety (IPCS, 2010) recommended that model predictions should typically be within a factor of two of the experimental data.

2.4. Local sensitivity analyses

Local sensitivity analyses were generally conducted by increasing a single input parameter value by 1%. The change in the prediction (e.g., plasma PFHxS concentration) was normalized to the concentration predicted using the baseline values in the model. Thus, the normalized sensitivity coefficient (NSC) was computed as percent change in output/percent change input and is dimensionless. A NSC value of "1" indicates a linear relationship between the input and output parameters, under the baseline conditions of the analysis, and that an increase in the input value produced an increase in the numerical value of the output. Negative values indicate that an increase in the input value produced a decrease in the value of the output. A low or zero value of NSC indicates a lack of sensitivity of the output to the input under the tested scenario.

The one-at-a-time sensitivity analysis approach for plasma PFHxS concentrations was applied separately to predictions for both men and women, multiple calendar years, and at ages at the approximate midpoints of ages used for evaluation of the NHANES data used for model calibration (CDC 2019, 2020). Additional sensitivity analyses for both plasma PFHxS concentration and the rate of urinary excretion of PFHxS were conducted for exposure scenarios yielding the median plasma concentrations for men and women in Fu et al. (2016).

3. Results and discussion

3.1. Scoping phase and initial model refinement and evaluation

The key findings of the scoping phase and the initial stages of model refinement and evaluation were reported in detail in the Supplementary Materials (additional text, Figure S-1 and Tables S-1 to S-4). Those findings are briefly summarized here. Two published human PBPK models of PFHxS were identified (Kim et al., 2018; Fàbrega et al., 2015). The model structures were very similar, consisting of a plasma compartment and distribution of unbound compound via plasma flow into several tissue compartments, and clearance of the chemical by way of renal filtration of unbound compound, with potential reuptake into the kidney. A lifestage PBPK model for PFOS/PFOA with similar pharmacokinetic elements and exposure descriptions incorporating calendar-year and age-dependent features was selected as the base model (Ruark et al., 2017) and further modified for PFHxS (depicted in Figure S-1 in the Supplementary Material). Gender-specific calibration

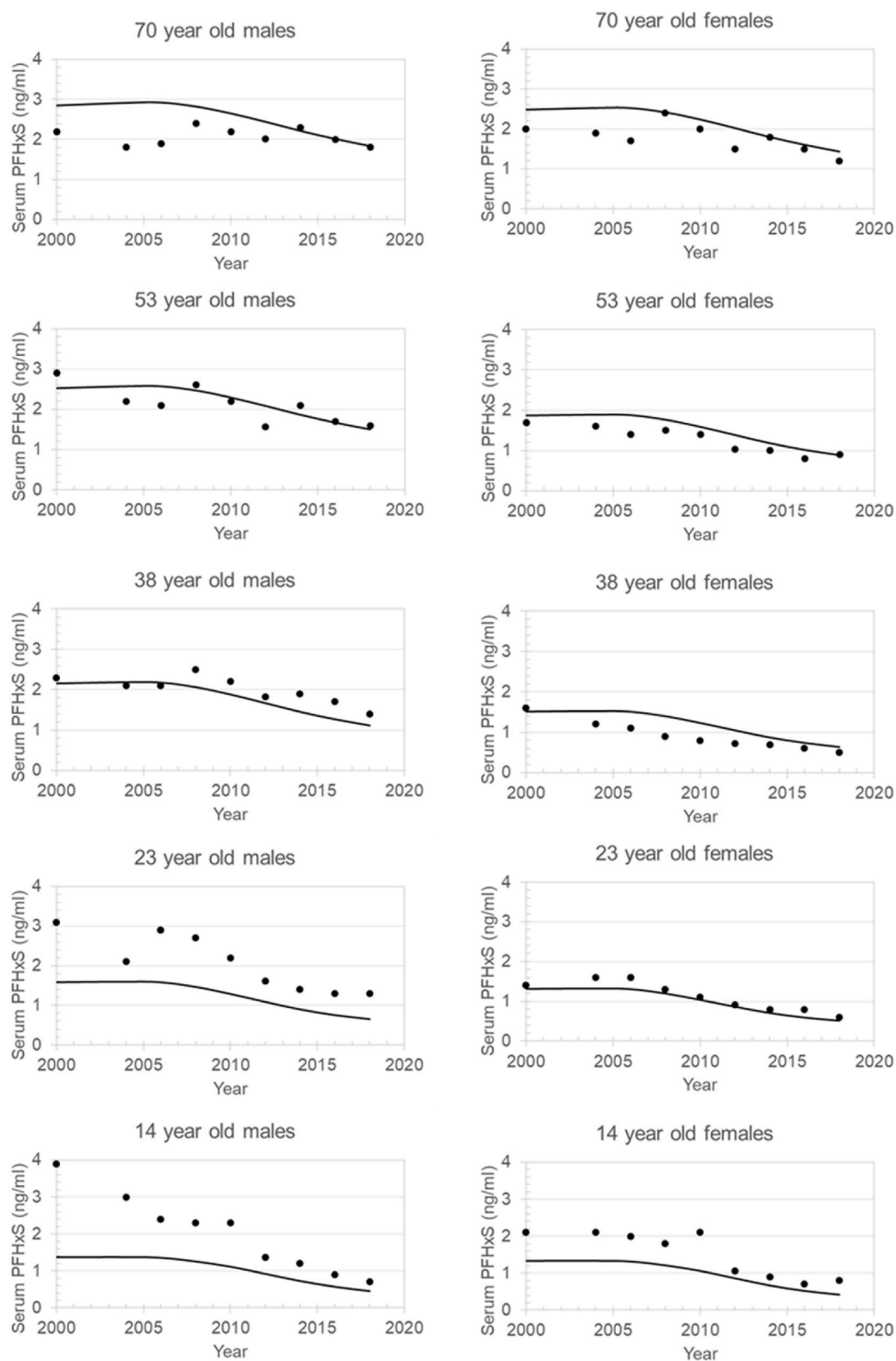


Fig. 1. Comparisons of model predictions to PFHxS NHANES data (CDC 2019, 2020). Line: model predictions for age group midpoints. Symbols: weighted median values. Age groups for CDC measurements were (top to bottom) age 61 years and up, ages 46–60 years, 31–45 years, 16–30 years, and 12–15 years. $n = 65$ to 363 per gender, age group, and year (see Supplementary Material Tables S–4 for details). Peak daily intake rate was a factor of 2.3 higher than that of Fàbrega et al. (2015).

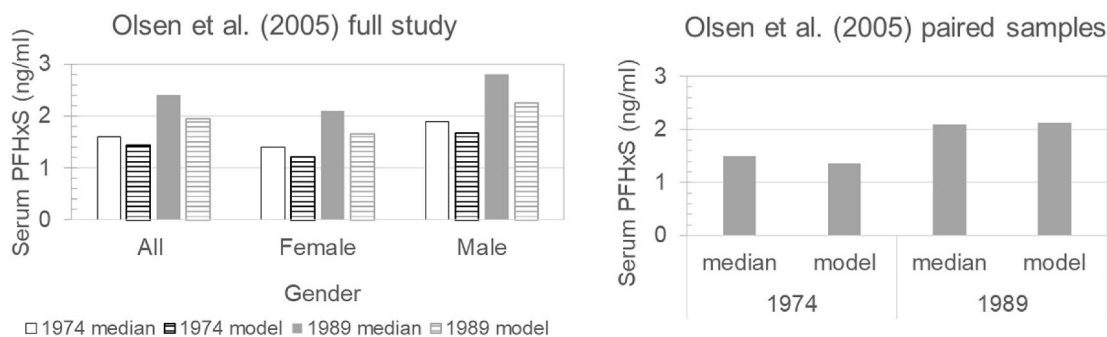


Fig. 2. Calendar year dependent serum PFHxS in adults residing near Hagerstown, Maryland. Median data from Olsen et al. (2005). Gender and calendar year specific model predictions were made for the reported central tendency ages, with a peak daily intake rate equivalent to that estimated for NHANES data (Fig. 1). Full study (left): 91 females and 87 males in 1974 (median age 47.5), 92 females and 86 males in 1989 (median age 52). Paired samples of individuals who participated in both 1974 and 1989 (right): $n = 58$; mean age 42 years in 1974, 57 years in 1989.

of PFHxS resorption relied on the data of Li et al. (2018). U.S. population, calendar year, and lifestage-specific intake calibration relied primarily on NHANES data (CDC 2019, 2020), with data from Olsen et al. (2005) used to calibrate longer-term historical trends (i.e., rate of exposure increase from 1950 to 1990). The resulting model performance as evaluated by comparison to the calibration data sets and other data was depicted in Figs. 1 and 2 and summarized in Table 1. Additional simulations and characteristics of studies noted in Table 1 may be found in the Supplementary Material (Figs. S-2 to S-8, Tables S-2 to S-4). Parameter values specific to the PFHxS model are summarized in Tables S-6 and S-7.

The agreement between the model and the calibration data was excellent, and recapitulated observed trends across sex, age, and calendar years. The greatest deviation between the model and the NHANES data (Fig. 1) was for younger males for 2000–2010. Plasma PFHxS concentrations for 5–14 and 15–24 year olds from Norway for a similar time period (1998–2007) do not show such marked male-female differences (Haug et al., 2009). The Olsen et al. (2017) data for 20–30 year old American Red Cross blood donors shows a modest male-female difference in 2000 (male:female concentration ratio of 1.2), but a 1.9 fold higher level in men of that age group in 2006 (Figure S-4). No explanation for the interstudy or regional difference in male:female plasma concentration ratios has been identified. With the exception of renal reabsorption of PFHxS, only exposure parameter values were optimized. When assessing agreement between the model predictions and the U.S. evaluation sets (Table 1), optimized intake rate for American Red Cross blood donors (Olsen et al., 2017) was only 13% less than the rate for NHANES participants. The optimized intake rate for California mothers of young children was about half the optimized NHANES rate, which may reflect regional or demographic differences. If no optimization of intake was included for the U.S.-based serum PFHxS evaluation studies, the model performance would still have been judged to be very good. In particular, the fit to data for eight consecutive years in the Kim et al. (2020) study suggests that sustained differences in exposure exist between this population and the NHANES populations, since adjustment of one exposure descriptor corrects the fit for the entire time span (Figure S-5).

The model generally overpredicted urinary excretion of PFHxS (Table 1, Figure S-8) by a factor of approximately 2- to 4-fold across a wide range of serum PFHxS concentrations (1.2–19,837 ng/ml). In general, PBPK models for PFOA and PFOS have not evaluated or presented the performance of the models with respect to human urinary data (Loccisano et al., 2011; Chou and Lin 2019, 2020; Worley et al., 2017) despite successfully reproducing serum data. It is not clear if the current discrepancy is due to differing relative amounts of clearance and reuptake in the kidney vs. gastrointestinal tract that are specific to PFHxS or if the other PFAS models would show similar discrepancies if they were likewise evaluated against the relatively recent urinary data

used in this analysis. Cholestyramine has previously been associated with enhanced elimination of PFAS, including PFHxS via the fecal route (Ducatman et al., 2021; Genuis et al., 2013).

3.2. Sensitivity analysis

Local sensitivity analyses were conducted that evaluated changes in predicted serum PFHxS concentrations for U.S. males and females based on small changes in assumptions about values of model input parameters. The findings reported in Table 2 demonstrate the complexity of the trends, with distinctly different patterns in sensitivity across combinations of gender, age, and calendar year. For example, assuming a slower rate decrease in background PFHxS exposure takes years to have a substantial impact on serum PFHxS levels, but has a greater impact on younger people. Likewise, an increase in the exposure multiplier for ages 0–10 years has a much greater impact on predicted serum PFHxS in 14-year-olds than 23-year-olds, but almost no impact on adults 38 years and older. The partitioning of PFHxS into the “rest of the body” is not a highly sensitive determinant of serum PFHxS, but the age and calendar year patterns of sensitivity indicate the changing dynamics of PFHxS distribution within the body. At younger ages and in the early 2000s, tissues served as a “sink” that lowered serum PFHxS. In contrast, in adults in later years, after environmental levels of PFHxS started to decline, storage tissues became a source for serum PFHxS. Predictions of serum PFHxS have complicated, gender-specific relationships to measures of body size (height and weight) because of the manner in which the Ruark et al. (2017) lifestage model is constructed. Physiological and anatomical parameters in the model are dependent on age and gender, and female body weights incorporate a correlation between weight and body mass index that is not present in calculations for males. Thus, scaling relationships for processes related to these variables differ between men and women. Sensitivity analyses for the 5-year occupational scenario of Fu et al. (2016) indicate that serum and urine concentrations in that investigation tend to be sensitive to the same chemical-specific, physiological, and anatomical parameters as the NHANES scenario, but to a differing degree. In the simulations of exposures monitored by Fu et al. (2016), serum and plasma concentrations show greater sensitivity to partition coefficients (fat and “the rest of the body”), likely because the exposure duration was insufficient to fully equilibrate to a level substantially above background; instead, these tissues serve as a sink for PFHxS. A complete reporting of the local sensitivity analyses is provided in the Supplementary Material (Tables S-8 to 10).

3.3. Model application

3.3.1. Comparison to predictions from toxicity reference value derivations with other modeling approaches

At least two toxicity reference value proposals have used

Table 1

Agreement between modeled and measured PFHxS serum or plasma concentrations in biomonitoring studies.

Data set	Model performance						Notes
	All subjects		All males		All females		
	AFE	AAFE	AFE	AAFE	AFE	AAFE	
U.S. calibration data sets (plasma)							
CDC 2019, 2020 (all)	0.904	1.32	0.810	1.38	1.01	1.27	Fig. 1. NHANES data (1999–2018), ages 12 y and older Background exposure multiplier and 2005–2013 exposure decline rate optimized
CDC 2019, 2020 age 16+	1.00	1.25	0.895	1.28	1.13	1.21	
Olsen et al., (2005) (all)	0.855	1.17	0.838	1.19	0.827	1.21	Fig. 2. Hagerstown, MD area adults (1974, 1989) 1950–1990 exposure increase optimized Paired samples were from individuals who participated in both blood collection years
Olsen et al., (2005) (paired samples)	0.962	1.04	Data not available				
U.S. evaluation data sets (plasma)							
Olsen et al. (2017)	1.03	1.37	0.978	1.37	1.08	1.37	Figure S-4. American Red Cross Data (2000–2015) Background exposure multiplier was optimized CDC 2019, 2020 background exposure multiplier was assumed
Kim et al. (2020)	1.18	1.41	1.12	1.39	1.24	1.43	
	1.03	1.04	None		Same as all		Figure S-5. Northern California mothers of a 2–5 year old child (2009–2016) Background exposure multiplier was optimized CDC 2019, 2020 background exposure multiplier was assumed
Hurley et al. (2018)	1.98	1.98	None		Same as all		
	1.01	1.03	None		Same as all		Figure S-5. California Teachers Study (2011–12 and -2013-2014) Background exposure multiplier was optimized CDC 2019, 2020 background exposure multiplier was assumed
Wang et al., (2011) (all)	1.45	1.45	None		Same as all		
	0.843	1.39	None		Same as all		One-time samples from three separate studies of California women CDC 2019, 2020 background exposure multiplier was assumed for all three studies.
Wang et al., (2011) (1960s)	1.04	1.72	None		Same as all		
	0.472	2.12	None		Same as all		Pregnant women (1960–1963). CDC 2019, 2020 background exposure multiplier was assumed.
Wang et al., (2011) (1980s)	1.26	1.26	None		Same as all		
Wang et al., (2011) (2009)	1.0	1.0	None		Same as all		Negative controls in a breast cancer study (1981–1986) CDC 2019, 2020 background exposure multiplier was assumed. Pilot study (2009) for the California Environmental Contaminant Biomonitoring Program Background exposure multiplier from Kim et al., (2020) applied CDC 2019, 2020 background exposure multiplier was assumed.
	1.92	1.92	None		Same as all		
Olsen et al. (2007)	1.003	1.07	1.003	1.07	1.003	1.07	Figure S-7, Tables S–6. Retired fluorochemical workers (22 male, 2 female) Data were used to characterize interindividual variability of maximal transporter mediated renal uptake of PFHxS Occupational exposure and renal reuptake rate were optimized
European calibration and evaluation data (plasma or milk)							
Haug et al. (2009)	1.01	1.26	Same as all		None		Figure S-6. Norwegian men age 40–50 y Background exposure multiplier and 1950–1990 exposure increase rate were optimized
Axmon et al. (2014)	1.03	1.34	None		Same as all		
Sundström et al. (2011)	0.972	1.30	None		Same as all		Figure S-6. Swedish women ~ age 48 y. Background exposure multiplier was optimized Figure S-6. Lactating Swedish women (~age 29 y) Plasma concentrations were estimated from pooled milk samples Background exposure multiplier was optimized
Sundström et al. (2011)	0.707	1.44	None		Same as all		
							Figure S-6. Lactating Swedish women (~age 29 y) Plasma concentrations were estimated from pooled milk samples Background exposure multiplier of Axmon et al. (2014) was applied
Evaluation of urine concentration (predicted from serum or plasma concentration)							
Fu et al. (2016)	2.34	2.61	2.20	2.59	2.64	2.67	Figure S-8. Chemical plant employees in China General population in two Chinese cities. Median serum and urine concentrations for full population (not segregated by gender). Fold error estimated from weighted average of simulations for males and females.
Zhang et al. (2013)	3.75	3.75	Not available		Not available		
Xu et al. (2020)	4.29	4.29	Not available		Not available		Workers at a PFAS-contaminated Swedish airport. Median serum and urine concentrations for full population; gender-weighted average for fold error estimates.

AFE = average fold error (Fold error = model prediction/measured value); value of 1 for AFE indicates balanced model performance. AAFE = average absolute fold error (maximum of fold error or 1/fold error); value of 1 for AAFE indicated perfect agreement between model and data. Additional study details are provided in the Supplementary Material, Tables S-2 and S-3.

pharmacokinetic modeling approaches to derive recommendations for PFHxS, either alone or as a PFAS mixture. Results of the PFHxS-specific PBPK model described here were compared to those prior efforts.

EFSA EFSA (2020) estimated that long-term maternal exposure to a sum of less than 0.63 ng/kg body weight/day of PFHxS, PFOS, PFOA, and perfluorononanoic acid (PFNA) would protect from adverse effects in humans. This external exposure value was derived by using a PBPK model for PFOS to estimate infants' intake of PFOS via breast feeding in the first year of life. Specifically, the EFSA model predicted that long term PFAS intake at this rate would result in a maternal serum level of 6.9 ng/ml at age 35 years. The use of the PFHxS-specific model reported here (which does not account for physiological changes associated with pregnancy) generated similar predictions for women: a level of 6.0 ng/ml at age 35 years, and a level of 7.1 at age 50 years.

In deriving a proposed chronic reference dose for PFHxS

reproductive toxicity based on experimental results in mice, Ali et al. (2019) used a one-compartment model of PFHxS kinetics. They used this one compartment model to convert a benchmark dose (95% lower confidence limit) of 13.9 µg/ml serum PFHxS to a human equivalent dose of 1195 ng/kg body weight/day. PBPK model predictions of the same intake rate for women were similar to the results of the one-compartment model, with estimates of 10.8 and 12.7 µg/ml serum PFHxS at ages 35 and 50 years, respectively.

3.3.2. Time to target predictions

In the absence of a specific problem formulation for a risk assessment, the “time to target” approach inverts the toxicity reference value derivation process. That is, the process starts with a target internal dose, and estimates additional (non-background) dose and duration combinations that could correspond to that internal dose for the simulated

Table 2

Normalized sensitivity coefficients for changes in serum PFHxS at specified years and ages for changes in select parameters.

Half-life of exposure decrease for 2005-13 (female)						Intake multiplier for ages 0-10 y (female)					
y\age	14	23	38	53	70	y\age	14	23	38	53	70
2000	0	0	0	0	0	2000	0.55	0.10	0.007	-0.001	0
2006	0.013	0.013	0.011	0.0099	0.007	2006	0.56	0.11	0.008	-0.001	0
2012	0.36	0.34	0.30	0.25	0.18	2012	0.72	0.17	0.012	0	0
2018	1.07	0.76	0.70	0.58	0.41	2018	0.65	0.28	0.026	0.0021	0.0014
Richly perfused tissue (rest of body):plasma partition coefficient (female)						Maximum transporter mediated renal resorption rate (female)					
y\age	14	23	38	53	70	y\age	14	23	38	53	70
2000	-0.12	-0.058	-0.049	-0.073	-0.081	2000	0.67	0.65	0.64	0.64	0.78
2006	-0.12	-0.042	-0.037	-0.060	-0.068	2006	0.68	0.67	0.66	0.66	0.81
2012	-0.026	0.086	0.083	0.043	0.011	2012	0.86	0.89	0.87	0.85	0.99
2018	-0.026	0.182	0.195	0.155	0.109	2018	0.87	1.05	1.05	1.04	1.21
Maximum height (female)						Maximum height (male)					
y\age	14	23	38	53	70	y\age	14	23	38	53	70
2000	-1.43	-1.20	-1.14	-0.05	-0.05	2000	-0.58	-0.53	-0.61	-0.65	-0.68
2006	-1.42	-1.18	-1.12	-0.05	-0.03	2006	-0.59	-0.55	-0.63	-0.67	-0.72
2012	-1.35	-0.97	-0.89	0.06	0.06	2012	-0.72	-0.71	-0.78	-0.82	-0.86
2018	-1.35	-0.86	-0.69	0.09	0.18	2018	-0.74	-0.85	-0.95	-1.00	-1.04
Maximum adult weight increase (female)						Maximum adult weight increase (male)					
y\age	14	23	38	53	70	y\age	14	23	38	53	70
2000	0.98	0.66	0.48	0.29	0.23	2000	0.82	0.80	0.47	0.29	0.24
2006	0.98	0.68	0.50	0.30	0.25	2006	0.81	0.80	0.49	0.31	0.26
2012	0.90	0.89	0.65	0.44	0.34	2012	0.57	0.97	0.60	0.42	0.36
2018	0.84	1.00	0.79	0.61	0.46	2018	0.60	0.90	0.75	0.57	0.47

Shading indicates parameter combinations with similar sensitivity within each block; cutoffs for shading vary among blocks.

Table 3

Predicted years of exposure until serum PFHxS exceeds 5 ng/ml, after start of exposure additional to background in 2022.

Background description ^a	Year of Birth	Predicted serum PFHxS in 2022 (ng/ml)	Other exposure (ng/kg bodyweight/d)					
			None	0.0333	0.1	0.333	1	3.33
<i>Females</i>								
Central tendency	2004	0.34	never	never	never	38.6	4.53	1.05
	1997	0.42	never	never	never	31.7	4.37	1.02
	1987	0.48	never	never	never	23.0	4.05	0.991
	1977	0.57	never	never	never	17.2	3.80	0.964
	1967	0.76	never	never	never	15.2	3.32	0.902
High (5x)	1957	1.04	never	never	never	14.5	3.10	0.844
	2004	1.70	never	never	never	23.2	2.87	0.738
	1997	2.10	never	never	never	17.1	2.54	0.652
	1987	2.40	never	never	never	11.8	2.16	0.577
	1977	2.84	never	never	never	7.11	1.75	0.478
	1967	3.79	never	never	never	3.12	0.883	0.258
	1957	5.21	-1.50 ^b	-2.40	until >80 ^c	until >80	until >80	until >80
<i>Males</i>								
Central tendency	2004	0.38	never	never	never	21.3	3.95	1.02
	1997	0.57	never	never	never	18.2	3.65	0.957
	1987	0.80	never	never	never	15.2	3.35	0.900
	1977	1.05	never	never	never	13.3	3.09	0.844
	1967	1.27	never	never	never	12.1	2.89	0.795
High (5x)	1957	1.46	never	never	never	11.4	2.73	0.755
	2004	1.90	never	never	never	8.95	2.45	0.681
	1997	2.84	never	never	never	6.30	1.64	0.468
	1987	4.02	never	never	28.1	2.79	0.747	0.214
	1977	5.23	-1.61	-2.48	until >80	until >80	until >80	until >80
	1967	6.35	-9.96	-16.2	until >80	until >80	until >80	until >80
	1957	7.30	until >80	until >80	until >80	until >80	until >80	until >80

^a Central tendency predictions use the same peak background exposure rate as predictions for NHANES studies (0.15 ng/kg body weight/d from 1990 to 2005; estimate of 0.036 ng/kg body weight/d in 2022).^b Negative values indicate years until model-predicted serum PFHxS declines below an initial 2022 concentration greater than 5 ng/ml.^c Model predictions were not extended beyond age 80 years due to limited calibration data.

subpopulations (Table 3). The target selected for this demonstration, 5 ng/ml, is only slightly lower than the internal target used by EFSA (2020) for PFAS including PFHxS, with no additional uncertainty (safety) factors applied in the assessment. The approach employed here considered men and women with baseline intake and an elevated intake that would put individuals into approximately the 95th percentile of the U.S. population. The central tendency background exposure estimate that served as a base line for the time-to-target analyses uses the calendar-year dependent time course derived for describing the NHANES data (Fig. 1). The estimated peak intake rate (for 1990–2005) was 0.15 ng/kg/d (a factor of 2.3 higher than the estimated rate for residents of Catalonia, used in the original Fàbrega et al. (2015) model). Based on a declining rate of intake (estimated for 2005–2013) and an assumption of a very slow continuing decline thereafter, a central tendency estimate intake of 0.036 ng/kg/d was projected for U.S. adults in 2022.

Males had shorter times-to-target and higher predicted PFHxS plasma levels due to higher rates for renal resorption in men (as determined by calibration to the data of Li et al., 2018; Supplementary Material Tables S–6) and greater PFHxS clearance for women during childbearing years (due to modeled menstrual loss). For individuals with typical historical exposures, exposure to around 10x the estimated current background for over a decade was predicted to be necessary to reach a serum PFHxS level of 5 ng/ml. However, those individuals who have had higher background levels that are consistent with NHANES 95th percentile data might already exceed the 5 ng/ml target at the beginning of the hypothetical exposure period.

4. Conclusions

Confidence in the model is greatest for application to adults in the 2000–2018 time frame and for shorter term future projections. U.S. data for periods prior to the earliest NHANES data were very limited. The 1974 and 1989 data from the Hagerstown area (Olsen et al., 2005, Fig. 2) were chosen for parameterizing changes in historic U.S. exposure (1950–1990) because the 1989 data appeared consistent with ~1999–2000 NHANES data. The NHANES predictions (2000–2018) have limited sensitivity to the 1950–1990 intake estimates (maximum NSC of 0.058 for 70-year-old men in 2000; Tables S–7). The 1974 data of Olsen et al. (2005) predate the earliest data of Haug et al. (2009) from Norway by three years. Nonetheless, those U.S. men (median population age: 47.5 y) in Olsen et al. (2005) had a median PFHxS serum concentration of 1.9 ng/ml, much higher than the levels of 0.1 and 0.29 ng/ml reported for 1977 and 1980 samples, respectively, for men ages 40–50 in Norway (Haug et al., 2009). Proper calibration of past exposure trends will improve internal dosimetry estimates for historical evaluation of health risks. The central tendency value for the U.S. PFHxS peak intake rate (1990–2005) estimated for the NHANES population was 0.15 ng/kg bodyweight/day, which is in the range of lower bound estimates from recent European food consumption surveys (0.09–0.86 ng/kg body weight per day EFSA (2020)).

As noted above, the only optimized toxicokinetic parameter values were renal resorption capacities for PFHxS. The requirement for differing optimized values for males and females (210 and 175 $\mu\text{g}/\text{h}/\text{kg}^{0.7}$) based on the data of Li et al. (2018) suggest that male-vs. female-specific differences (e.g., body composition and growth) and menstrual blood loss were insufficient to account for observed male-female differences in serum PFHxS. The current model does not account for maternal PFHxS losses due to childbirth, breastfeeding, or other toxicokinetic alterations during the perinatal period. While Jain (2013) did not find a statistically significant association between parity and plasma PFHxS in 17–39 year old women in 2003–2008 NHANES studies, other studies of U.S. women have made adjustments to serum PFHxS predictions based on parity (Mondal et al., 2014; Hu et al., 2019; Kim et al., 2020). Recent NHANES data also support that pregnancy is associated with lower plasma PFHxS levels (Tables S–5). Individually optimized

renal resorption capacities for 22 male fluorochemical industry retirees (Tables S–7, based on data of Olsen et al. (2007), 141 ± 58 [mean $\pm$ SD], range 49–272 $\mu\text{g}/\text{h}/\text{kg}^{0.7}$) were generally lower than the values derived from the Li et al. (2018) data for 15–50 year-olds.

The similarity between predictions using the PFHxS PBPK model and other approaches (EFSA (2020), Ali et al., 2019) provides additional support for the parameterization and structure. The time-to-target simulations demonstrate particular considerations involved in assessing the acceptability of a given intake level for a variable population. Key differences include gender-related clearance capacity, interindividual differences in physiology and exposure, and calendar-year dependence of exposures to this family of persistent compounds. Physiologically-based models have the capacity to address many of these limitations; improved exposure characterization and continued biomonitoring will be needed for model refinement to better address future risk assessment needs. The current model provides additional flexibility to incorporate historic exposure trends, as compared to the Ruark et al. (2017) model for PFOS and PFOA. In contrast to the two previously published PFHxS models (Fàbrega et al., 2015; Kim et al., 2018), this PFHxS model has been more rigorously compared to population time course plasma data plus it has been compared to urinary data; recapitulated trends for calendar year, age and gender; and model code has been provided (Supplementary Material) for transparency. When specific PFHxS risk assessment applications prompt considerations of PBPK models, variability analyses, global or targeted sensitivity analyses, or development of models focused on particular life stages (the very young, pregnant women, or lactating mothers) could potentially be valuable extensions of the current work.

Disclaimers

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Declaration of competing interest

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Appendix A. Supplementary data

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